Does the Early-Warning System Work? Prediction, Causation and Targeting in a Synthetic Student-Retention Cohort

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Deliverable 3: Results and Discussion

Everything in this deck is synthetic.

Every number describes a generative model calibrated to cited aggregate statistics.
It is not a measurement of any real student, institution or programme.

I use the data like an analyst, and the hidden truth only to grade my work

The setup

- A synthetic cohort of **50,000 students** over 8 terms, calibrated to cited Philippine attrition: **67.1%** (high tier) and **45.9%** (typical tier).
- A **hidden** state (baseline vulnerability, acute shocks) drives the records *and* the dropouts. An analyst sees only eight exported tables.
- Support is assigned from prior-term signals, so it is **confounded**. The true effect is known: the dropout hazard is cut by **30%**.

The rules I held myself to

- Analysis reads the **exported CSVs only**. The hidden truth is used in three labelled places, always to *grade* the result.
- **No leakage**: a feature for term t uses entry attributes and terms 1 to $t-1$ only.
- Validation **grouped by student**; same folds for every model; nothing tuned on test data.
- Every added choice tagged `LOCKED`, `CALIBRATED` or `ASSUMED`.

Because I know the answer, I can say whether each method gets the *right* answer, not just whether it produces one.

Six questions, one chain of evidence

1. Is the generator valid?

7 of 7 re-derived checks match the scorecard. The engine reproduces all 8 tables bit for bit.

2. What does the data show?

Attrition 67.1% vs 45.9% by tier, highest for low-income working students.

3. Can I predict dropout?

AUC **0.655** vs 0.567 for today's rule. Ceiling with perfect knowledge: 0.722.

4. Does support work?

The raw data say it **harms** (−58%). Done right: **+32%** against a true 30%.

5. Who is missed?

64% of dropouts occur with a clean record and no support.

6. How robust is this?

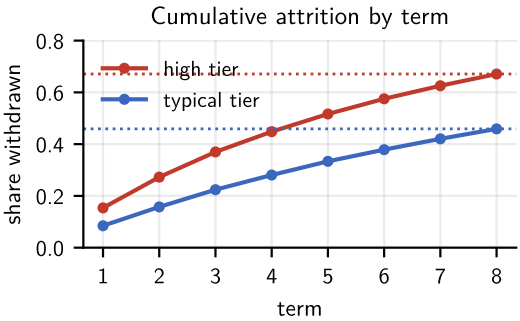
Holds across 11 recalibrated scenarios. IPW fails without overlap.

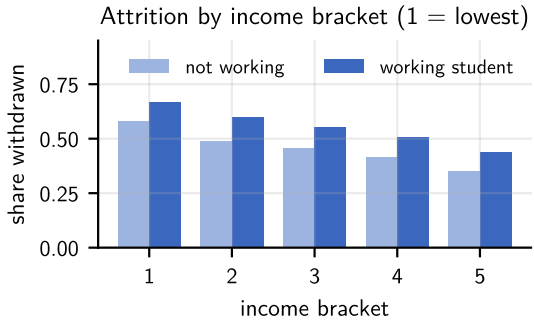
The generator holds up when I re-derive its checks from the CSVs

Check	Re-derived from the exported tables	Result
Tier attrition (calibrated)	High 67.1%, typical 45.9% (targets 67.1%, 45.9%)	pass
Correlation structure	Signals correlate (+0.39, −0.19, −0.18), but ≤ 0.009 once the hidden state is removed	pass
Confounding present	Naive estimate −57.7% (95% CI −64.6 to −50.6) excludes the true 30%	pass
Effect recoverable	Mantel–Haenszel 32.2% (24.3 to 39.3); Poisson 33.6% (25.1 to 41.2)	pass
Edge cases from the mechanism	2,937 unflagged shock-driven dropouts; 774 falsely flagged terms	pass
Engine copy is faithful	All 8 tables regenerate with matching SHA-256; calibration loop returns the recorded values	pass

What this does not show. It does not show the generator resembles any real institution. Calibration to the two tier targets is a convergence check, not independent validation. I did not re-run the generator's held-out Class-of-2023 check, which it labels *pass (fragile)*.

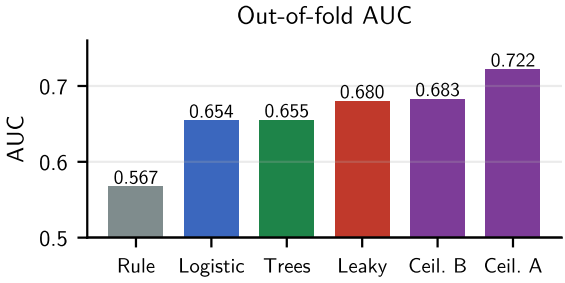
What the data shows: two tiers, a steep gradient, and a selection effect

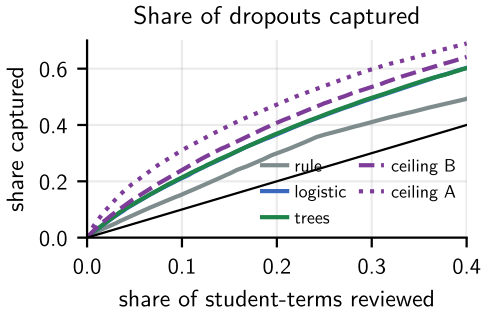




- Per-term dropout *falls* over time (high tier 15.4% to 12.1%): that is **selection**, since the most vulnerable leave first. It is not evidence that retention gets easier.
- Attrition runs from about **35%** (non-working, top income bracket) to **67%** (working, lowest). The gradient is built into the generator: a structure check, not a finding about the world.

Early warning works, but only modestly





- **Boosted trees: AUC 0.655** (fold SD 0.005) against 0.567 for the signal-count rule. A logistic hazard gets 0.654, so the signal is nearly additive.
- Reviewing the top **10%** of student-terms captures **21.4%** of dropouts (rule: 15.3%); calibration is good in every decile.
- Drivers: the *running averages* of prior grades and attendance, not last term's flags.

How good could any model be? The hidden truth sets the ceiling

Model	AUC	PR-AUC	Top 5%	Top 10%	Top 20%
Signal-count rule	0.567	0.104	8.2%	15.3%	29.9%
Logistic hazard	0.654	0.139	12.1%	21.1%	36.7%
Boosted trees	0.655	0.140	12.2%	21.4%	36.9%
Leaky: + current-term signals (not deployable)	0.680	0.163	14.6%	24.8%	40.4%
Ceiling B: knows baseline vulnerability	0.683	0.159	13.9%	24.2%	40.8%
Ceiling A: true hazard	0.722	0.219	19.7%	31.1%	47.4%

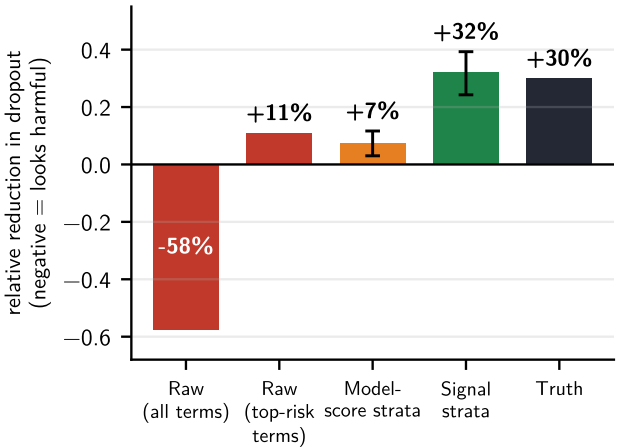
Out-of-fold, 247,739 student-terms (terms 2 to 8), event rate 8.0%. "Top x%" = share of dropouts captured by reviewing the top x% of student-terms.

Much of dropout cannot be foreseen. Even the true hazard gives AUC 0.722, because this term's shock is unknowable in advance.

19.4% of dropouts (4,762 of 24,573) happen in term 1, before any record exists. No term-based model can see them.

Leaving out regional tier costs nothing (AUC 0.655 vs 0.656), so excluding it on ethical grounds has no accuracy price here.

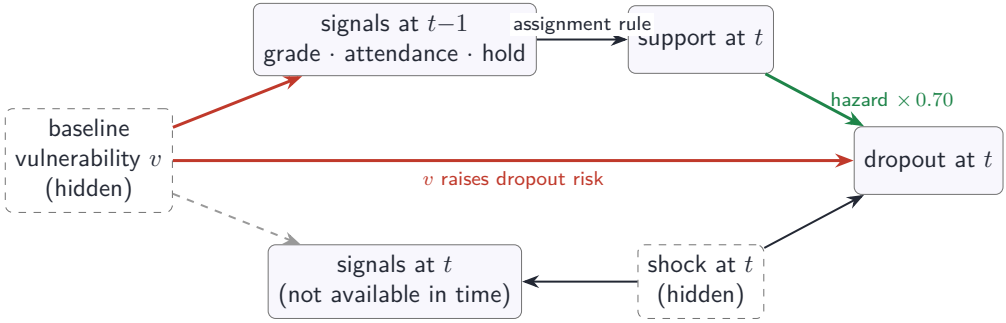
A good predictor cannot tell me whether support works



- **Raw comparison: −58%.** Support goes to the students most at risk, so it looks harmful (confounding by indication).
- **Stratify on the model's score:** +7% (CI +3.0 to +11.7). Right sign, but it misses the truth.
- **Stratify on the assignment signals:** +32.2% (CI +24.3 to +39.3). Covers the true 30%.

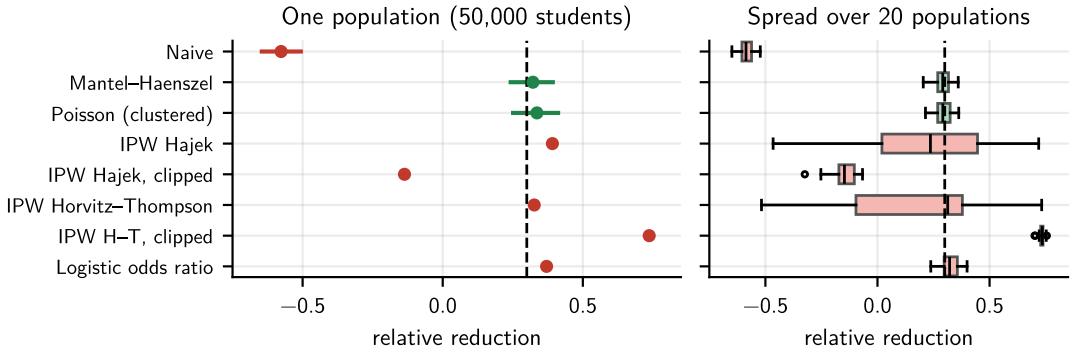
Ranking students and evaluating a programme need **different designs**.

Why: a hidden common cause opens a backdoor path



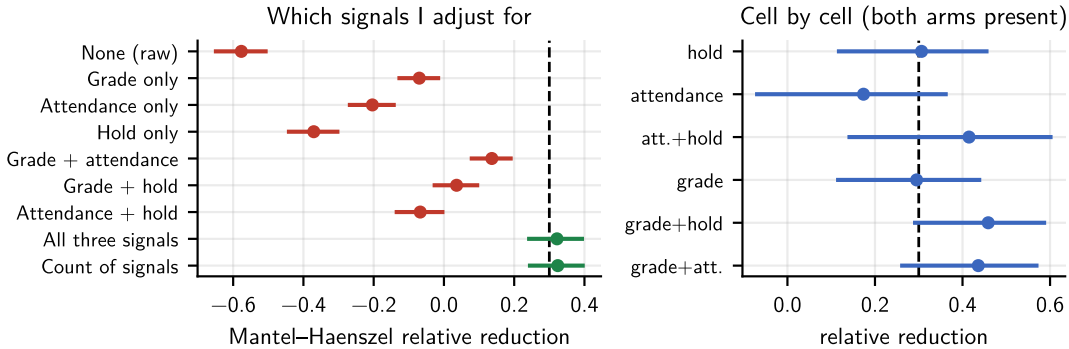
- The path support $\leftarrow$ signals$(t-1)$ $\leftarrow v \rightarrow$ dropout is open, which is why the raw comparison fails.
- Support depends *only* on the signals at $t-1$, so within a signal cell it is as good as random: **stratify on those signals**.
- The shock at t is independent of everything before it, so it cannot confound.

Estimator choice matters more than sample size



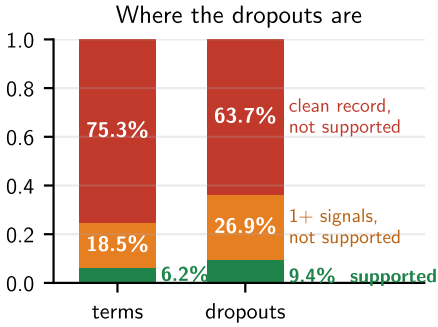
- **Mantel–Haenszel and Poisson** recover about 30% (SD about 4 points over 20 populations): one study resolves the effect to roughly ±8 points.
- **IPW swings from −47% to +72%** over the same populations: the supported arm's *effective* sample size is **43** of 15,293 (and I gave IPW the true propensity).

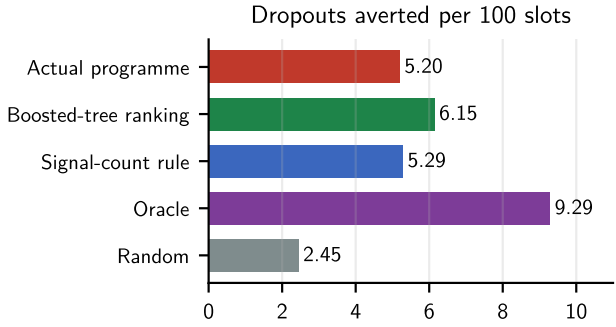
Adjusting for only some of the referral signals fails



- Grade only −7.0%, attendance only −20.4%, hold only −37.1%. **Only the full set (+32.2%) or the count of signals (+32.4%) recovers the effect.**
- **Log the inputs to the referral decision, not just the flag**, or the programme looks harmful however well it works. Only six cells hold both arms, so effect heterogeneity is out of reach.

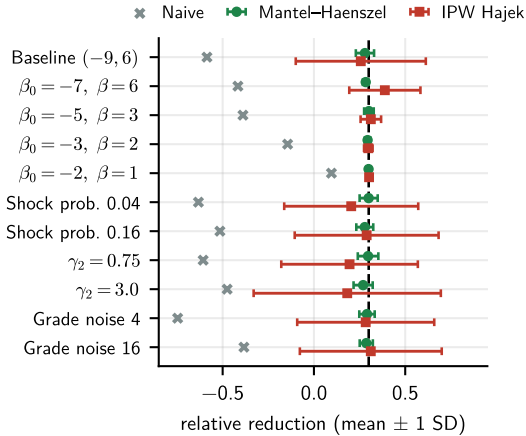
Most dropouts happen where the programme is not looking





- **63.7%** of dropout events fall in clean-record, unsupported terms; only 9.4% fall in supported terms (6.2% of all terms).
- At equal capacity (15,293 slots) the model ranking averts an expected **940** dropouts against **796** for the actual rule (+18%), but only 66% of an unattainable oracle (1,421). This assumes a homogeneous 30% effect.

The conclusions survive what I varied



- **11 scenarios**, each re-fitted to the same 67.1% / 45.9% targets, so differences are not just differences in attrition level.
- **Mantel–Haenszel stays near 30%** everywhere (replicate means 26.9% to 30.1%).
- **IPW instability is a positivity problem**: its SD falls from 35.6 points (default rule) to 1.1 (softest rule) as every signal cell gains both arms.
- Predictability is **capped by baseline vulnerability**: ceiling B stays within 0.659 to 0.686, while ceiling A ranges from 0.696 to 0.800 with shock strength.

Recommendations: how to build and evaluate a retention programme

1. **Triage, do not judge.** Use the risk model to allocate attention, never to evaluate the programme.
2. **Evaluate with the assignment rule.** Log the referral inputs with the support flag; use Mantel–Haenszel or clustered Poisson. Expect ±8 points at 50,000 students.
3. **Build overlap into the rollout.** A small randomised share of referrals makes IPW and cell estimates usable.
4. **Pilot model-based allocation** head to head with the current rule before trusting the +18%.
5. **Look where the dropouts are,** not only where the flags are (64% have a clean record).
6. **Keep regional tier out** of the flagging model (no accuracy cost here).
7. **Handle IPW with care** under steep assignment: report effective sample size and largest weights.
8. **Do not act on these numbers.** The method is the point: validate against known truth, separate prediction from causation, and test how much each conclusion depends on assumptions.

What these results are not

- **Everything is a property of the generator.** The prediction ceilings, the homogeneous 30% effect and the form of the confounding are assumptions, not measurements.
- **The causal problem here is easier than a real one.** Support never changes later grades (no carry-over), assignment has no unobserved drivers or caseworker discretion, the analyst knows the thresholds, and IPW gets the true propensity.
- **The targeting gain is conditional.** The policy value assumes the same 30% effect for everyone, so ranking by risk is optimal by construction.
- **The model comparison is narrow.** Two model families, fixed hyperparameters, one population; the sensitivity grid varies one parameter at a time with 10 replicates each.
- **Open items.** The generator's held-out check is fragile and was not re-run; the mechanism behind the biased score-stratified estimate is not investigated; term-1 dropouts (19.4%) fall outside both analyses.

Full list with status: notebook Section 12 (flags OF-1 to OF-12 from the generator, DF-1 to DF-9 new here).

Take-aways

1. A risk model **ranks** students. It does not tell me whether the support **works**.
2. Evaluating support needs the **assignment inputs**, the right adjustment set, and **overlap** between those who were and were not supported.
3. Inside this synthetic world a better ranking buys **modestly** more, because much of dropout is unforeseeable and most of it happens where no flag is raised.

Where to find it. Notebook: `results-and-discussion/student-retention-analytics.ipynb` (executed copy: `.html`). It runs end to end from the exported CSVs, and every output carries a synthetic-data banner. Dataset and generator: `synthetic-data-generator/`. Specification: whitepaper v17.

References

Gelman, A., Hill, J., and Vehtari, A. (2020). *Regression and Other Stories*. Cambridge University Press, Cambridge, U.K.

James, G., Witten, D., Hastie, T., and Tibshirani, R. (2013). *An Introduction to Statistical Learning with Applications in R*. Springer, New York, NY.

Law, A. M. (2015). *Simulation Modeling and Analysis*, 5th edition. McGraw-Hill, New York, NY.

Lopez-Muñoz, G. L., Gonzalez Serrano, C., and Sanchez-Ferreira, C. (2026). Student Dropout Prediction in Higher Education: A Systematic Review of Machine Learning Methods and Risk Factors. *Journal of Information Technology Education: Research*, 25, Article 20. https://doi.org/10.28945/5776

Peters, J., Janzing, D., and Schölkopf, B. (2017). *Elements of Causal Inference: Foundations and Learning Algorithms*. MIT Press, Cambridge, MA.

The generator's calibration sources (CPBRD 2025; EDCOM II 2025; CHED 2025) are traced in the whitepaper, Section 10.

Backup A: policy values at equal capacity

Policy (15,293 slots)	Averted	Per 100 slots	% of oracle	Slots per averted
Actual programme	796	5.20	56.0%	19.2
Boosted-tree ranking	940	6.15	66.2%	16.3
Signal-count rule, ties at random	809	5.29	56.9%	18.9
Oracle (true untreated hazard)	1,421	9.29	100.0%	10.8
Random	375	2.45	26.4%	40.8

Expected values inside the model: a slot is worth 0.30× the student-term's untreated hazard. Capacity is the number of supported student-terms in terms 2 to 8 (6.2% of 247,739). Total avertable if every term were supported: 6,097 dropouts.

- The oracle ranks by true untreated hazard, which includes *this term's shock*. That is why it is unattainable, and why a below-average-vulnerability student can still be its best pick.
- The signal-count rule with random ties performs like the actual rule, since the actual rule is that rule plus noise.

Backup B: sensitivity grid in numbers

Scenario	Supp.	Naive	MH	MH SD	IPW SD	AUC rule	AUC B/A
Baseline $(-9, 6)$	6.2%	−57.7%	+32.2%	5.0	35.6	0.565	0.683/0.722
$\beta_0 = -7, \beta = 6$	10.7%	−38.0%	+33.5%	1.5	19.5	0.558	0.680/0.720
$\beta_0 = -5, \beta = 3$	7.0%	−35.3%	+33.8%	2.6	5.6	0.566	0.684/0.723
$\beta_0 = -3, \beta = 2$	12.9%	−10.3%	+33.1%	1.5	2.3	0.563	0.683/0.723
$\beta_0 = -2, \beta = 1$	17.0%	+12.1%	+32.3%	1.0	1.1	0.570	0.686/0.726
Shock prob. 0.04	5.6%	−59.0%	+30.9%	5.0	36.7	0.564	0.684/0.707
Shock prob. 0.16	7.4%	−52.5%	+31.8%	4.6	39.4	0.562	0.679/0.743
$\gamma_2 = 0.75$	6.3%	−58.3%	+31.0%	5.6	37.4	0.565	0.686/0.696
$\gamma_2 = 3.0$	6.1%	−49.8%	+35.0%	5.2	51.3	0.556	0.659/0.800
Grade noise 4	4.9%	−74.9%	+32.2%	4.2	37.5	0.561	0.684/0.722
Grade noise 16	8.2%	−37.7%	+30.9%	3.6	38.9	0.557	0.682/0.722

Each scenario recalibrated to 67.1% / 45.9%. Estimates are relative reductions (truth +30%). "MH SD" and "IPW SD" are SDs in percentage points over 10 replicate populations. "Supp." is the share of student-terms (terms 2 to 8) that receive support.